MDPI

*Article*

# WRF Cold-Pool Structure and Evolution during a Uinta Basin Winter Ozone Episode

**John R. Lawson [1,2,*], Michael J. Davies [1], Loknath Dhar [1,3] and Seth N. Lyman [1,3]**

[1] Bingham Research Center, Utah State University, Vernal, UT 84078, USA;
[2] Department of Mathematics and Statistics, Utah State University, Logan, UT 84322, USA;
[3] Department of Chemistry and Biochemistry, Utah State University, Logan, UT 84322, USA
* Correspondence: john.lawson@usu.edu

**Abstract**

Cold-air-pool structure sets the mixing volume and transport wind that a winter-ozone chemistry model inherits from its meteorological driver, and surface skill alone does not establish it. We compared four deterministic Weather Research and Forecasting (WRF) simulations of the 2 February 2013 Uinta Basin episode, crossing GFS and NAM driving products with one-way and two-way nest feedback at eleven fixed 1 km station columns over five post-spin-up hours (14:00–18:00 UTC). Relative to NAM, GFS increased the paired 2 m-temperature-bias contrast by 1.79 and 1.76 °C under two-way and one-way feedback, reduced 50–500 m stability by 3.63 and 3.65 K, and increased fixed-layer transport wind by 0.79 and 0.99 m $s^{-1}$, with the same sign in all five hours. Over a registered evaluation footprint, the GFS members also carried 0.70 to 1.80 MJ $m^{-2}$ less integrated heat deficit than the feedback-matched NAM members at every retained hour under seven footprint definitions. Heat deficit nevertheless increased in every member, so a predeclared test for faster GFS heat-deficit loss failed in all fourteen pair-definition combinations. Surface-pressure errors ordered the products oppositely, and within-product feedback effects were much smaller throughout. The experiment does not separate initial from lateral-boundary forcing, validate inversion depth, establish multi-day persistence, or predict ozone.

**Keywords:** winter ozone; WRF; cold-air pool; Uinta Basin; driving data; nest feedback; heat deficit; model evaluation

## 1. Introduction

Winter ozone in the Uinta Basin develops when photochemistry proceeds within snow-covered, stably stratified air that restricts vertical dilution and horizontal exchange [1–3]. Snow increases actinic flux and modifies deposition, while the stable, weakly ventilated air confines ozone and its precursors near the basin floor. For a coupled model, the meteorological question is therefore not only whether 2 m temperature has a small mean error. It is whether the atmosphere delivered to the chemistry grid contains a credible shallow mixing environment and weak transport during the accumulation period.

Published concentration agreement does not provide a clean ranking of the relevant uncertainties. Ahmadov et al. obtained a much stronger ozone simulation with a top-down oil-and-gas inventory, while also conditioning inventory interpretation on adequate meteorology [1]. Matichuk et al. found substantial emissions and deposition sensitivities, but improved WRF temperature profiles did not consistently improve ozone profiles [4]. These outcomes are consistent with compensating error among confinement, emissions,

Received:
Revised:
Accepted:
Published:

**Copyright:** © 2026 by the authors. Submitted to *Air* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

photolysis, deposition, and chemical mechanism. They make meteorological evaluation necessary for chemical attribution without making meteorology the uniquely dominant error source.

The meteorological literature also warns against reducing an experiment to a configuration label. Neemann et al. showed strong case-specific responses to snow and cloud treatment, but their ice-retention modification was a bounding treatment and the grid evaluated most closely was finer than the grid supplied to the chemistry model [2]. Tran et al. showed that the response to four-dimensional data assimilation depended on the nudged variables, input data, and diagnosed boundary-layer depth [5]. Thus, labels such as "NAM driven," "nudged," or "high resolution" do not by themselves identify a controlled cause, and aggregate scores can obscure a failure to retain the layer of interest.

The present study asks three deliberately bounded questions. First, does the short-window lower-layer state differ between GFS- and NAM-driven WRF simulations under both nest-feedback modes? Second, how large is the within-product nest-feedback response relative to that driving-product contrast? Third, does the driving-product response also appear in a footprint-integrated heat-deficit measure of the simulated cold pool, and do the products differ in how that measure changes across the window? We answer these questions with paired surface errors, common-height stability and wind diagnostics, and an area-weighted heat-deficit integral on one domain, station field, and five-hour window. The heat-deficit comparison carries two criteria that were fixed before any output was inspected, and we report the outcome of both. The design evaluates the meteorological input state that a later chemistry calculation would receive; it does not contain a chemistry calculation.

## 2. Materials and Methods

### *2.1. Factorial Design and Estimands*

The experiment crossed the archived GFS analysis product `gfsanl_4` and NAM analysis product `namanl_218` with one-way (`feedback=0`) and two-way (`feedback=1`) nest feedback (Table 1). Stable study codes connect each member to the run and comparison inventories. The products supplied zero-hour analyses at 12:00 and 18:00 UTC and provided 27 and 40 metgrid levels for GFS and NAM, respectively. The GFS-versus-NAM factor is therefore a product-level treatment: it bundles the products' native grids, analysis systems, vertical input, assimilated observations, 12:00 UTC initial state, and subsequent lateral-boundary evolution. The experiment cannot assign the resulting response to one of those components. Within each product family, the normalized as-run namelists differed only in the feedback flag. The two product contrasts were evaluated separately so that the driving-product response had to recur under both feedback settings to support the primary result. The two within-product feedback contrasts and their paired interaction bounded the secondary response.

**Table 1.** Driving-product by nest-feedback experiment matrix. The study codes resolve to exact run paths and fingerprints in the run inventory.

| Study code | Driving-data product | Nest feedback | Manuscript label |
|---|---|---|---|
| X6 | GFS | two-way | GFS two-way |
| X10 | GFS | one-way | GFS one-way |
| X5 | NAM | two-way | NAM two-way |
| X3 | NAM | one-way | NAM one-way |

### 2.2. Case, Observations, and Common Sample

The target is the snow-covered Uinta Basin cold-air pool on 2 February 2013 within the episode examined by prior meteorological and chemical studies [1,2]. All factorial comparisons use 14:00–18:00 UTC, corresponding to 07:00–11:00 mountain standard time. The first two forecast hours were excluded as spin-up, leaving five hourly samples; the sensitivity of the results to that two-hour choice has not been evaluated. Surface observations were extracted from the Synoptic archive for eleven stations reporting 2 m temperature, ten reporting 10 m wind speed, and nine reporting surface pressure in every retained hour. Subhourly reports were averaged within each UTC clock hour. The stations span basin-floor, sidewall, and bench locations but are neither spatially uniform nor area weighted. The 1 km second domain, d02, was selected because the 333 m third domain excludes most higher bench stations. Four elevation-spanning stations provide surface context from 31 January 00:00 UTC through 4 February 00:00 UTC. Observed and modeled surface potential temperature for a two-site structural index was calculated as $\theta = T(1000/p)^{0.2854}$, with $T$ in kelvin and $p$ in hPa, and the index was defined as Mountain Home on the northwest bench minus Pariette Draw on the basin floor. Because the two sites are horizontally separated and locally exposed, the index is a spatial surface contrast rather than a colocated free-air inversion strength or depth. No sounding, profiler, or remote-sensing profile is used to validate the modeled 50–500 m layer.

**[Rewrite gate: Document the Synoptic citation and access date, API quality-control parameters, station ownership, sensor heights and metadata, completeness screening, observation uncertainty, public-access route, and archive license.]**

### 2.3. WRF Configuration

The four free forecasts used WRF version 4.8.0 from 12:00 to 18:00 UTC on 2 February 2013. Three nested domains had horizontal grid spacing of 3, 1, and 0.333 km, 75 eta levels, and a 50 hPa model top. The common physics suite used Thompson microphysics, RRTMG longwave and shortwave radiation, the revised MM5 surface layer, the Noah land-surface model, and the YSU planetary-boundary-layer scheme. Grid and observation four-dimensional data assimilation were disabled. The runs shared domain geometry, default 30-arc-second-class terrain, physics, initialization time, output cadence, and valid window. The GFS one-way member reused the six verified metgrid files from the GFS two-way member before rerunning `real.exe` and `wrf.exe`.

**[Rewrite gate: Add the WRF and exact GFS/NAM product citations, map projection, domain dimensions and centers, time step, boundary interval, static-geography versions, and snow-field initialization from the archived as-run records.]**

### 2.4. Surface and Common-AGL Diagnostics

Modeled 2 m temperature, earth-relative 10 m wind, and surface pressure were sampled at the nearest d02 cell to each station. Across the eleven fixed columns, model terrain minus station elevation had a median of -2.0 m and ranged from -32.4 to +14.3 m. That agreement limits one obvious mismatch but does not remove exposure, slope, land-cover, or network-coverage error. Mass-level height above ground was calculated from perturbation and base-state geopotential and the local WRF terrain height. Grid-relative winds were destaggered and rotated to earth-relative components. Potential temperature and both wind components were interpolated linearly at 25 m increments from 50 to 500 m above ground level (AGL) at each station column. Fixed-layer stability was defined as $S_{50:500} = \theta(500\,\mathrm{m}) - \theta(50\,\mathrm{m})$. The transport-wind proxy was $U_{50:500} = \{\bar{u}^2 + \bar{v}^2\}^{1/2}$, where the bars denote height means from 50 to 500 m AGL. This wind metric is not flow through the basin boundary, pollutant residence time, cold-pool volume, or a chemistry-model

ventilation coefficient. Native WRF planetary-boundary-layer (PBL) height was retained only as a scheme-dependent companion diagnostic.

### *2.5. Footprint-Integrated Heat Deficit*

The column heat deficit was defined as $H = (c_p/g) \int \max\{\theta_{\text{crest}} - \theta, 0\}\, dp$, integrated from the surface to a fixed crest reference height, where $\theta_{\text{crest}}$ is the potential temperature interpolated to that height in each column. **[Rewrite gate: Add the heat-deficit method citation (Whiteman-style valley heat deficit).]** $H$ measures the energy required to remove the sub-crest stable layer and is reported as a cell-area-weighted mean over an evaluation footprint in MJ $m^{-2}$. The footprint is the convex hull of the eleven evaluation stations and eleven registered basin waypoints, intersected with model terrain below the crest reference. The primary definition uses a 2200 m crest and the combined hull, covering 4965 d02 cells and 5277 $km^2$. Six alternate definitions vary the crest to 2100 and 2300 m, dilate or erode the boundary by one cell, or restrict the hull to the stations or waypoints alone. Because two-way feedback modifies the d02 terrain field by up to 23.1 m relative to the static one-way field, differences were formed within feedback mode, X6 minus X5 and X10 minus X3, on a common mask fixed from the one-way NAM terrain; the members of each pair share identical grids. Hourly values at 13:00 UTC provide trajectory context only, and all decisions use 14:00–18:00 UTC. Two criteria were fixed before inspection: a weaker-state criterion requiring both GFS-minus-NAM pairs to be negative at every retained hour under all seven definitions, and a faster-loss criterion requiring both GFS 14:00-to-18:00 UTC changes to be negative and more negative than the matched NAM changes under all seven definitions. The footprint is an evaluation region, not a hydrologic basin boundary, and the integral is not an observed cold-pool measurement.

### *2.6. Pairing, Aggregation, and Short-Sample Scope*

Run differences were paired by station and valid hour before spatial aggregation. The forcing-by-feedback interaction was defined as $[\text{GFS}_{1\text{w}} - \text{GFS}_{2\text{w}}] - [\text{NAM}_{1\text{w}} - \text{NAM}_{2\text{w}}]$ and was likewise evaluated at each station and hour before spatial aggregation. Each station field was reduced to one unweighted spatial median per hour, and the five hourly medians were averaged for the reported effect. The stations were not treated as independent replicates. Hourly ranges and sign counts describe temporal consistency directly. A two-hour circular moving-block resample of the five hourly medians, repeated 20,000 times, and leave-one-hour-out ranges were retained in the reproducibility record as descriptive sensitivity checks. They are not confidence intervals, hypothesis tests, or estimates of observation, forcing, or structural uncertainty. The heat-deficit criteria in Section 2.5 were likewise fixed before any model output was inspected. No population inference is attempted from four deterministic integrations of one case.

## 3. Results

### *3.1. Observed Surface Context*

The basin-floor station at Pariette Draw repeatedly cooled below the higher sites from 31 January through 3 February, while the magnitude and ordering of the station temperatures changed through the diurnal cycle (Figure 1). The recurring floor-to-bench separation is consistent with cold-pool conditions around the modeled period. It does not determine the cold-pool top, depth, volume, or vertical temperature structure.

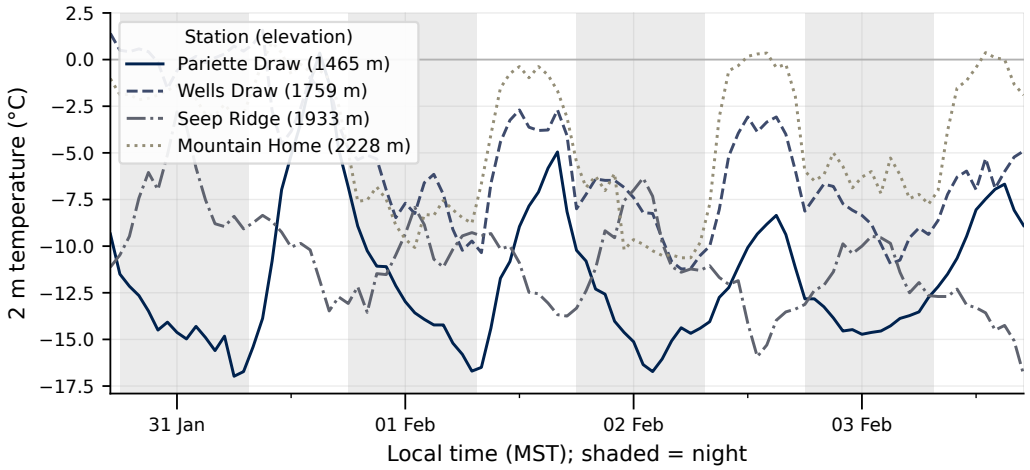



**Figure 1.** Observed 2 m-temperature evolution at four elevation-spanning Uinta Basin stations from 31 January through 3 February 2013. Shading indicates approximate local night. The sites establish a recurring spatial surface-temperature pattern but do not constitute a colocated vertical profile.

### *3.2. Driving-Product Response in the Common Lower Layer*

The common-AGL diagnostics separated the driving products under both feedback settings (Figure 2). GFS minus NAM reduced 50–500 m potential-temperature stability by 3.63 K under two-way feedback and 3.65 K under one-way feedback (Table 2). The five hourly spatial-median differences ranged from -5.58 to -1.13 K and from -5.36 to -0.90 K, respectively. All ten product-by-hour stability contrasts therefore retained the same sign. GFS minus NAM increased fixed-layer transport wind by 0.79 m $s^{-1}$ under two-way feedback and 0.99 m $s^{-1}$ under one-way feedback. The hourly differences ranged from +0.37 to +1.18 m $s^{-1}$ and from +0.61 to +1.50 m $s^{-1}$, with the same sign in every hour. Native PBL-height differences changed sign within both feedback modes and did not reproduce the stability and wind separation.

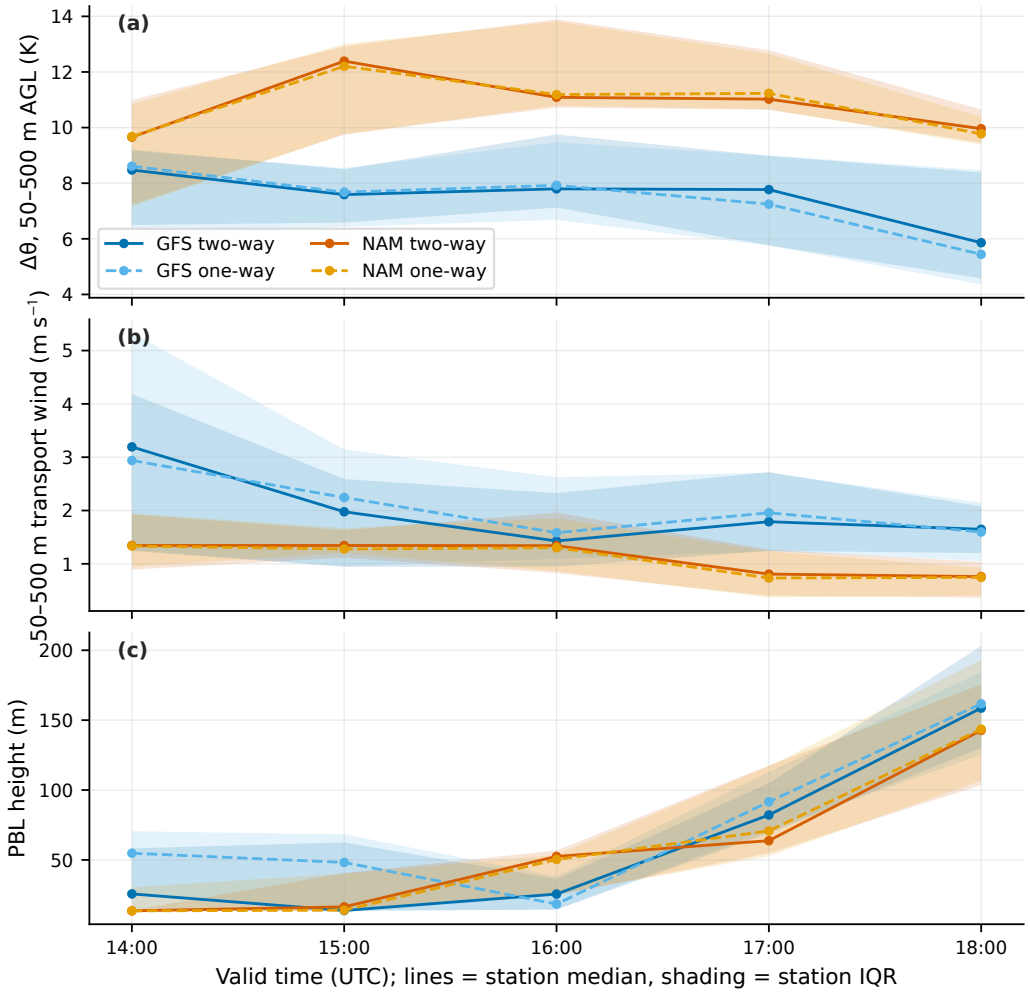



**Figure 2.** Station-column medians and interquartile ranges for 50–500 m AGL potential-temperature stability, fixed-layer transport wind, and native WRF PBL height over the matched window. Solid lines denote two-way feedback and dashed lines denote one-way feedback. The lower-layer GFS–NAM separation recurs under both feedback modes, while PBL height is retained as a scheme-dependent companion rather than an observed layer depth.

**Table 2.** Paired factorial effects averaged over the five hourly spatial medians. Positive temperature values indicate a warmer model-minus-observation bias, positive stability values indicate a more stable fixed layer, and positive transport-wind values indicate a larger layer-mean wind. The effects are deterministic descriptive contrasts rather than inferential estimates.

| Effect | ΔT2 bias (°C) | $\Delta S_{50:500}$ (K) | $\Delta U_{50:500}$ (m s$^{-1}$) |
|---|---|---|---|
| GFS − NAM, two-way | +1.79 | -3.63 | +0.79 |
| GFS − NAM, one-way | +1.76 | -3.65 | +0.99 |
| GFS one-way − two-way | +0.07 | -0.07 | +0.10 |
| NAM one-way − two-way | -0.01 | +0.00 | +0.01 |
| Feedback interaction | +0.04 | -0.10 | +0.10 |

### *3.3. Surface Corroboration and Metric-Dependent Skill*

The paired surface-temperature result had the same product ordering as the layer diagnostics. Mean hourly station-median 2 m-temperature biases were +2.09 and +2.51 °C for the GFS members, against +0.40 and +0.60 °C for the matched NAM members. After pairing by station and hour, GFS minus NAM increased the temperature-bias contrast by +1.79 °C under two-way feedback and +1.76 °C under one-way feedback, and the

corresponding absolute-error differences were +1.41 and +1.50 °C (Table 2). GFS minus NAM also increased 10 m-wind bias by +0.91 and +0.96 m $s^{-1}$, although the hourly difference in absolute wind error changed sign. Surface pressure ordered the products in the opposite direction. GFS minus NAM changed the paired surface-pressure absolute error by -13.6 Pa under two-way feedback and -10.9 Pa under one-way feedback, negative in all five hours, while all four members retained a positive mean pressure bias between +43 and +56 Pa on the nine-station sample. These pressure differences are small in absolute terms, but their consistent sign shows that the surface metrics do not agree on a product ranking. Aggregate surface skill is therefore metric-dependent for these simulations, and it cannot by itself establish which member supplies the more credible lower layer.

The two-site potential-temperature index provides the only direct observation-model comparison of lower-layer structure in this study (Figure 3). Under two-way feedback, the GFS-driven member had a smaller Mountain Home-minus-Pariette Draw contrast than the matched NAM-driven member in every retained hour. The observed spatial contrast ranged from 12.63 to 20.21 K, the NAM member from 16.44 to 18.07 K, and the GFS member from 7.35 to 11.74 K. These two-site values combine elevation, horizontal distance, site exposure, and local surface energy balance, and they are not interpreted as free-air inversion depths.

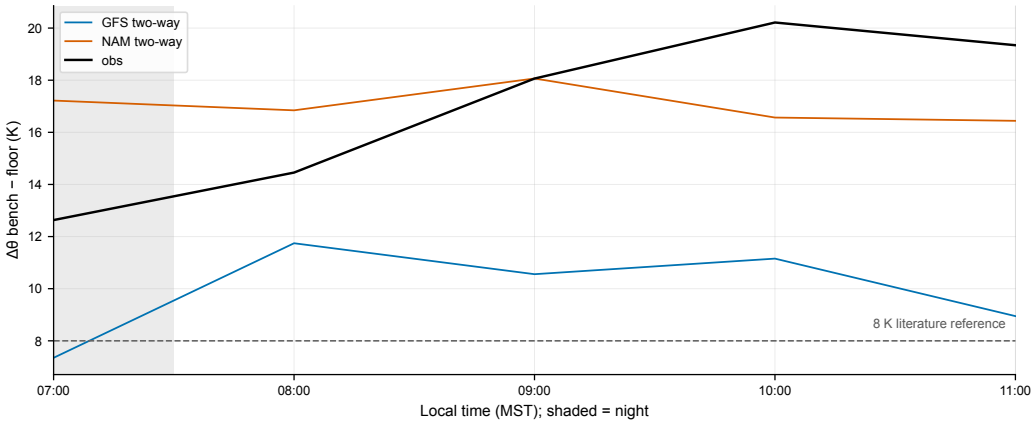



**Figure 3.** Observed and modeled Mountain Home-minus-Pariette Draw potential-temperature contrast during the matched evaluation window. Only the two-way members are shown. The two sites are horizontally separated surface stations, so the contrast is a structural index rather than a free-air inversion depth. The plotted 8 K line is a literature reference value, not a case-specific classification threshold. **[Rewrite gate: Add the 8 K reference source citation before submission.]**

### *3.4. Footprint Heat Deficit: A Weaker State without Faster Loss*

Under the primary definition, the footprint-mean heat deficit of the GFS members rose from 2.68 to 3.00 MJ $m^{-2}$ (two-way) and from 2.66 to 2.95 MJ $m^{-2}$ (one-way) between 14:00 and 18:00 UTC, while the matched NAM members rose from 4.18 to 4.33 and from 4.19 to 4.32 MJ $m^{-2}$ (Table 3). The paired GFS-minus-NAM differences averaged -1.23 MJ $m^{-2}$ under two-way feedback and -1.25 MJ $m^{-2}$ under one-way feedback. All 70 retained pair-hour-definition differences were negative, spanning -1.80 to -0.70 MJ $m^{-2}$, so the weaker-state criterion passed in all 14 pair-definition combinations. The faster-loss criterion required an actual GFS decrease that was also larger than the matched NAM change. Instead, heat deficit increased in every member, by +0.32 and +0.30 MJ $m^{-2}$ for GFS and by +0.15 and +0.13 MJ $m^{-2}$ for NAM under two-way and one-way feedback. The difference in change was +0.17 MJ $m^{-2}$ under both feedback modes for the primary definition and spanned -0.07 to +0.32 MJ $m^{-2}$ across definitions, without a consistent sign. The faster-loss criterion therefore failed in all 14 pair-definition combinations. The

GFS-minus-NAM separation consequently narrowed slightly over the window, from -1.50 to -1.33 MJ $m^{-2}$ under two-way feedback and from -1.53 to -1.36 MJ $m^{-2}$ under one-way feedback. The footprint result shows a consistently weaker simulated pool in the GFS members at every retained hour, not a divergence in its evolution.

**Table 3.** Footprint-integrated heat-deficit contrasts over the 14:00–18:00 UTC decision window. The primary definition uses the combined station-and-waypoint hull with a 2200 m crest (4965 cells, 5277 $km^2$); envelopes span all seven footprint definitions. The values are deterministic descriptive contrasts in MJ $m^{-2}$.

| GFS − NAM quantity | Two-way pair | One-way pair |
|---|---|---|
| Mean difference (primary) | -1.23 | -1.25 |
| Hourly range (primary) | -1.50 to -0.97 | -1.53 to -0.99 |
| Hourly envelope (seven definitions) | -1.77 to -0.70 | -1.80 to -0.71 |
| GFS change, 14:00–18:00 UTC | +0.32 | +0.30 |
| NAM change, 14:00–18:00 UTC | +0.15 | +0.13 |
| Difference in change (primary) | +0.17 | +0.17 |
| Weaker-state criterion passed | 7/7 | 7/7 |
| Faster-loss criterion passed | 0/7 | 0/7 |

### *3.5. The Nest-Feedback Bound*

The within-product feedback responses were much smaller than the GFS-minus-NAM contrasts, but they were not identical between products (Table 2). For NAM, one-way minus two-way averaged +0.002 K for stability and +0.006 m $s^{-1}$ for transport wind, and the hourly effects changed sign within -0.042 to +0.088 K and -0.035 to +0.033 m $s^{-1}$. For GFS, one-way minus two-way averaged -0.066 K for stability and +0.104 m $s^{-1}$ for transport wind. The GFS stability effect ranged from -0.198 to +0.067 K and was negative in four of five hours, while the transport-wind effect ranged from +0.025 to +0.197 m $s^{-1}$ and was positive in all five hours. The paired feedback interaction averaged -0.103 K for stability and +0.097 m $s^{-1}$ for transport wind, with the wind component positive in all five hours. The 2 m-temperature-bias interaction was +0.041 °C, and the PBL-height interaction changed sign from hour to hour. These values bound the response of the present configuration and sample; they do not establish general equivalence between feedback modes.

## 4. Discussion

### *4.1. A Product-Level Difference, Not a Universal Product Ranking*

The principal result is a replicated product-level separation. The GFS-driven members are warmer at the evaluated surface sites, less stable across the fixed 50–500 m layer, faster in the fixed-layer transport metric, and lower in footprint-integrated heat deficit than the NAM-driven members under both feedback settings. Replication across the feedback flag makes this ordering insensitive to that one configuration choice within the present experiment. It does not establish that NAM is generally superior, and the opposite ordering of the surface-pressure errors shows that even the present surface evaluation does not support a single-metric ranking. The product factor combines the 12:00 UTC initial atmosphere with boundary updates through 18:00 UTC. A direct comparison of each product with an independent profile at initialization would test inherited lower-tropospheric error, and an experiment that holds either the initial or boundary state fixed would be needed to apportion the subsequent response. Until those tests exist, "driving-product response" is more accurate than "initial-condition effect."

### 4.2. What the Difference Means for an Air-Quality Model

The GFS members supplied a less stable, faster-moving, and lower-heat-deficit lower layer over the evaluated columns and footprint. That combination is consistent with a meteorological state more favorable to dilution than the matched NAM state. The distinction is relevant because a chemistry calculation cannot recover a shallow accumulation volume that is absent from its meteorological input. The footprint heat deficit extends the fixed-column result to an area integral of the simulated pool: the GFS members carried 0.70 to 1.80 MJ $m^{-2}$ less deficit at every retained hour under every footprint definition. However, the transport-wind metric does not measure pollutant export, the heat-deficit footprint is not an observed volume, and no chemical species were simulated. This boundary is important in the Uinta Basin literature. Inventory changes, snow-enhanced photolysis, deposition, chemical mechanisms, and meteorological confinement have all produced substantial, interacting responses [1,2,4]. An improved meteorological state need not yield improved ozone if another component changes or compensates. The present result therefore identifies a meteorological precondition for attribution rather than assigning an ozone effect or ranking all sources of model error.

### 4.3. A Predeclared Heat-Deficit Test and Its Negative Result

The heat-deficit analysis carried two criteria that were fixed before any output was inspected. The weaker-state criterion passed unanimously and extends the fixed-column stability result to an area-integrated measure of the simulated pool. The faster-loss criterion failed unanimously because the heat deficit of every member increased through the morning window. We report that outcome rather than relaxing the criterion after inspection. The correct reading is a state offset that was already present at the first retained hour, not a demonstrated within-window divergence in pool evolution between the products. A hypothesis that the GFS members progressively deplete the simulated pool is not supported on this window. Testing differential heat-deficit loss at all would require a longer matched integration through at least one diurnal cycle, with the same predeclared footprint and criteria.

### 4.4. Why the Common-AGL Result Leads the Paper

This draft gives the common-AGL panel priority over both the surface scores and the two-site index. The surface metrics disagree with one another: temperature favors NAM strongly, pressure favors GFS consistently but weakly, and the wind-error ordering depends on the error definition. A single aggregate surface score would therefore conceal exactly the layer behavior this experiment was designed to expose. The two-site index is useful descriptive context and supplies the paper's only direct observation-model comparison, but its stations are horizontally separated, only its two-way product pair is shown, and it cannot validate a free-air inversion. By contrast, the common-AGL calculation applies the same geometric layer, fixed columns, valid times, and aggregation to all four simulations and directly displays both experimental factors. Native PBL height is not a replacement for that calculation: its product contrasts changed sign, and its physical meaning depends on the boundary-layer scheme. This behavior follows the broader lesson from data-assimilation and cold-pool studies that diagnosed PBL depth, nudging reach, and evaluation-grid structure must be disclosed rather than treated as interchangeable [2,5].

### 4.5. Limits That Determine the Next Experiment

The five-hour window is the strongest limit. It can show a coherent short-window separation but cannot establish multi-day retention, event buildup, or cold-pool breakup, and the heat-deficit trajectories cover the same five hours. The two-hour spin-up exclusion

has not been varied, and resampling five dependent hourly medians cannot create an inferential sample. The observation network creates a second limit. Four stations establish surface context and eleven fixed columns support matched comparisons, but neither set observes the modeled vertical layer, and small nearest-cell elevation errors do not remove exposure and representativeness error. Independent in-basin vertical observations are required to evaluate inversion depth, layer stability, and the simulated heat deficit directly. Observation provenance and the full model-configuration record must also be completed before submission, as marked by the rewrite gates in this draft. Four archived controlled questions remain outside this paper under predeclared gates: the spatial and vertical coherence of the driving-product response, a terrain-geometry comparison, one slope-shading or boundary-layer process comparison, and the feedback response over fine terrain. Each enters a future revision only if its analysis passes the corresponding gate, and none is presented as a result here. Only after those steps should selected meteorological states be crossed with emissions and chemical mechanisms in a coupled attribution experiment.

## 5. Conclusions

Four matched WRF simulations of the 2 February 2013 Uinta Basin episode show a coherent five-hour response to the driving-data product on the common d02 evaluation sample. Relative to NAM, GFS is warmer at the evaluated surface sites, less stable from 50 to 500 m AGL, faster in the fixed-layer transport-wind metric, and 0.70 to 1.80 MJ $m^{-2}$ lower in footprint-integrated heat deficit at every retained hour, under both one-way and two-way nest feedback. Surface-pressure errors order the products in the opposite direction, so aggregate surface skill alone would not identify these lower-layer differences. A predeclared faster-loss criterion for the GFS heat deficit failed in all fourteen pair-definition combinations because the deficit increased in every member; the driving-product response is a state offset within this window, not a demonstrated difference in pool evolution. The within-product feedback responses are much smaller for all evaluated diagnostics. The result establishes a short-window difference in the meteorological state available to downstream chemistry; it does not identify the responsible product component, validate inversion depth, demonstrate event-scale persistence, calculate basin export, or predict ozone. The next discriminating steps are a longer matched factorial with an independent in-basin vertical reference and a design that separates initial from lateral-boundary contributions.

**Author Contributions: [Rewrite gate: Assign and approve CRediT roles for all authors.]**

**Funding: [Rewrite gate: Insert verified funders and grant numbers.]**

**Institutional Review Board Statement:** Not applicable.

**Informed Consent Statement:** Not applicable.

**Data Availability Statement:** The analysis scripts, derived hourly and paired summaries, heat-deficit tables, run and comparison inventories, provenance records, and promoted figures used for this draft are retained in the private working repository. **[Rewrite gate: Add a public repository or DOI, licenses and access conditions for raw WRF and Synoptic data, and the exact release commit before submission.]**

**Acknowledgments: [Rewrite gate: Identify computational facilities, data providers, and contributors who do not meet the authorship criteria.]**

**Conflicts of Interest: [Rewrite gate: All authors must confirm the conflicts-of-interest statement.]**

**Use of Artificial Intelligence:** Generative artificial intelligence tools, including Anthropic Claude and OpenAI Codex, were used during manuscript preparation to assist with literature synthesis,

provenance and analysis-code review, and drafting and editing. The authors reviewed and edited the resulting material and take responsibility for the content.

## References


1. Ahmadov, R.; McKeen, S.; Trainer, M.; Banta, R.; Brewer, A.; Brown, S.; Edwards, P.M.; De Gouw, J.A.; Frost, G.J.; Gilman, J.; et al. Understanding high wintertime ozone pollution events in an oil-and natural gas-producing region of the western US. *Atmos. Chem. Phys.* **2015**, *15*, 411–429.
2. Neemann, E.M.; Crosman, E.T.; Horel, J.D.; Avey, L. Simulations of a cold-air pool associated with elevated wintertime ozone in the Uintah Basin, Utah. *Atmos. Chem. Phys.* **2015**, *15*, 135–151. https://doi.org/10.5194/acp-15-135-2015.
3. Davies, M.J.; Lawson, J.R.; O'Neil, T.; Lyman, S.N.; Zager, K.; Coxson, T.D. Uinta Basin snow shadow: Impact of snow-depth variation on winter ozone formation. *Air (Basel)* **2025**, *3*, 22. https://doi.org/10.3390/air3030022.
4. Matichuk, R.; Tonnesen, G.; Luecken, D.; Gilliam, R.; Napelenok, S.L.; Baker, K.R.; Schwede, D.; Murphy, B.; Helmig, D.; Lyman, S.N.; et al. Evaluation of the Community Multiscale Air Quality Model for Simulating Winter Ozone Formation in the Uinta Basin. *J. Geophys. Res. D: Atmos.* **2017**, *122*, 13545–13572. https://doi.org/10.1002/2017JD027057.
5. Tran, T.; Tran, H.; Mansfield, M.; Lyman, S.; Crosman, E. Four dimensional data assimilation (FDDA) impacts on WRF performance in simulating inversion layer structure and distributions of CMAQ-simulated winter ozone concentrations in Uintah Basin. *Atmos. Environ. (1994)* **2018**, *177*, 75–92. https://doi.org/10.1016/j.atmosenv.2018.01.012.


**Disclaimer/Publisher's Note:** The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.